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PC-SD-VIO: A constant intensity semi-direct monocular visual-inertial odometry with online photometric calibration

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ABSTRACT

The brightness constancy assumption is the cornerstone of direct or semi-direct visual odometry (VO) and visual simultaneous localization and mapping (SLAM). However, due to the existence of automatic exposure time, nonlinear camera response function, and vignetting, this assumption is difficult to hold in practical applications. Therefore, the corresponding algorithm performs poorly on arbitrary video sequences and uncalibrated cameras. Hence, we propose a novel constant intensity semi-direct visual-inertial odometry (VIO) integrated with online photometric calibration, which combines the exactness of the feature-based method and the quickness of the direct method. We combine gain-adaptive direct image alignment and gain-adaptive Kanade–Lucas–Tomasi (KLT) optical flow tracking to complete the feature matching and use it as the input for the back-end optimization and online photometric calibration. Our photometric calibration module can complete the estimation of all photometric parameters without any prior knowledge and cooperate with the front-end to complete the real-time photometric calibration of the latest frame. Experiments on the TUM Mono VO dataset, EuRoC dataset, and real environments prove that the algorithm can reliably calibrate the photometric parameters of an arbitrary video sequence. The semi-direct VIO algorithm integrated with the photometric calibration algorithm achieves a good balance between speed, accuracy, and robustness.

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1. Introduction

Visual SLAM and VO algorithms can provide robots with accurate and robust state estimation (an important part of robotic autonomous systems) in GPS-denied environments [1–4]. In the past, many visual SLAM or VO methods based on direct, semi-direct, and optical flow methods have been proposed, such as DSO, LSD-SLAM, SVO, and VINS-Mono [5–8]. Since these algorithms all consider pixel intensity, they all rely on the brightness constancy assumption, where the projection of the same scene point on different images should have the same pixel intensity value. However, in practical applications, the brightness constancy assumption is usually invalid. To ensure that the video sequence or image captured in different lighting conditions and scenes is neither too dark nor too bright, the camera automatically adjusts the exposure time, thus breaking the brightness constancy assumption. Due to the existence of vignetting, the pixel intensity of the captured image usually decreases radially from the center of the image, which also breaks this assumption [9–12]. In addition, the nonlinearity of the camera response function causes the pixel intensity of the same scene point on different images to be different. Therefore, algorithms related to pixel intensity have difficulty achieving good performance in real environments.

In this paper, we integrate the online photometric calibration algorithm into a novel semi-direct visual-inertial odometry to solve the problem of inconstant photometry. Our algorithm combines gain-adaptive direct image alignment and gain-adaptive KLT optical flow tracking to obtain feature point correspondences at the front-end and uses these results as input for back-end optimization and photometric calibration to estimate the response function, vignetting, exposure time, camera pose, and scale. To summarize, the main contributions of our work are as follows:

- An online photometric calibration algorithm that can be integrated into the semi-direct VIO and the main criterion for determining whether the photometric calibration converges and terminates.
- A novel constant intensity semi-direct visual-inertial odometry. The tracking front-end can cooperate with the photometric calibration module to calibrate the original uncalibrated image, thereby eliminating the negative effects of the response function, exposure time, and vignetting on the brightness constancy assumption.

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2. Related works

In the past, many photometric calibration methods have been proposed to calibrate and eliminate vignetting and automatic exposure as well as solve nonlinear response functions. The photometric calibration algorithm proposed in [13–16] requires manual adjustment of some photometric parameters to complete the calibration of other photometric parameters. Some photometric calibration methods, such as those in [17,18], can only calibrate some photometric parameters. However, these algorithms cannot run on an arbitrary video sequence without providing them with some photometric calibration input.

Goldman et al. [19] modeled the photometric calibration problem as a nonlinear energy function and optimized the joint estimation of all photometric parameters through the Gauss–Newton algorithm. However, since this work focuses on image stitching, a large number of accurate correspondences are required and therefore cannot be applied to a system with high real-time requirements such as SLAM. Bergmann et al. [20] proposed an online photometric calibration algorithm for visual SLAM, which can significantly improve the performance of direct visual SLAM. However, the photometric calibration algorithm is not integrated into the SLAM algorithm, so it is necessary to process the uncalibrated video sequence before it can be applied to the SLAM algorithm.

At present, many SLAM or VO systems have been proposed, but few systems consider photometric calibration, such as VINS-Mono and SVO. For cameras without photometric calibration, DSO can estimate photometric parameters online while estimating camera motion [10]. DSO estimates the image correction parameters to roughly remove the influence of the response function and the exposure time but does not consider vignetting. To make the direct or semi-direct SLAM algorithm more competitive when applied to arbitrary video sequences and uncalibrated cameras, we propose integrating the online photometric calibration algorithm into the SLAM algorithm. It can calibrate all photometric parameters without any prior knowledge, thereby ensuring the validity of the brightness constancy assumption.

3. PC-SD-VIO system

3.1. System framework overview

An algorithm overview of the system is shown in Fig. 1. For each new uncalibrated image frame, the response function and vignetting estimated by online photometric calibration are first used to process the image frame to eliminate its influence on the image. If the latest frame is a non-keyframe, the feature points are tracked by gain-adaptive direct image alignment. If the latest frame is a keyframe, it is tracked by gain-adaptive KLT optical flow method [21].

To complete the accurate estimation of the vignetting and response function, a large number of uniformly distributed and radially moving feature points are required as inputs, which results in a slow optimization speed [20]. Considering that the exposure time can be estimated frame by frame, we decouple this estimation from other photometric parameters and estimate the exposure time immediately after tracking the latest frame. Finally, the latest frame is calibrated to eliminate the influence of exposure time, thereby completing the calibration of the latest image frame.

We optimize the vignetting and response function in the background so that it does not affect the real-time performance of the SLAM algorithm. If the latest frame is a keyframe, the calibrated keyframe and feature points are stored in the keyframe database. The photometric calibration module obtains optimized data from the keyframe database to complete the continuous update of the response function and vignetting. The successful calibration of the response function and the vignetting depends on whether there are enough uniformly distributed feature points and whether the feature points have sufficient radial motion. Therefore, by checking the distribution frequency of feature points on the image and the number of feature points of large radial motion in the keyframe database, we can determine whether the photometric calibration has converged and terminated.

After photometric calibration and feature point tracking, the visual-inertial alignment is completed by combining the IMU pre-integration information. Finally, at the back-end, we minimize the reprojection error of monocular vision, the IMU residual, and the prior information of marginalization to obtain accurate state estimation [22].

3.2. Online photometric calibration based on optimization

The camera captures the reflected light L of the spatial feature point (with Lambertian reflectance behavior) and outputs the light intensity O on the image under the influence of the exposure time e, the vignetting V, and the response function R.

$$O = R(eV(x)L) \tag{1}$$

We adopt the empiric model of response (EMoR) proposed by Grossberg and Nayar [8] to model the response function.

$$R = R_0 + \sum_{k=1}^{n} c_k h_k \tag{2}$$

This model has been successfully applied to the estimation of the response function in the panorama stitching system. We select the parameter c_k, estimate the average response function R_0 and the basis function h_k, and combine them linearly to form the global response function R.

We adopt the flexible radial vignetting model used in [6], assuming that the vignetting is symmetric around the image center and model the vignetting as a sixth-order polynomial.

$$V(x) = 1 + v_1 r(x)^2 + v_2 r(x)^4 + v_3 r(x)^6 \tag{3}$$

where $r(x)$ is the normalized radius of the image point x with respect to the image center.

Considering that the estimation of the vignetting and response function requires a large number of uniformly distributed and large radially moved feature points, this inevitably increases the computational cost. The SLAM system requires high real-time performance, so it is difficult to put the photometric calibration module directly into the SLAM system. Inspired by the work of [20], considering that the exposure time can be estimated frame by frame, we decouple the exposure time estimation from other photometric parameters. We place the estimator of the vignetting and response function into the background to run independently and estimate the exposure time after tracking the feature points. The slower estimated vignetting and response function does not affect the real-time performance of the SLAM system.

When the latest frame arrives, the image is first calibrated through the response function and vignetting calculated in the background, and then there is only the negative effect of the exposure time. The latest frames are tracked through gain-adaptive direct image alignment and gain-adaptive KLT optical flow. We have improved the standard version of direct image alignment and KLT optical flow tracking so that it can track feature points and estimate the exposure time ratio between adjacent frames at the same time. In this way, the tracking of feature points is not affected by the exposure time. Although the exposure time

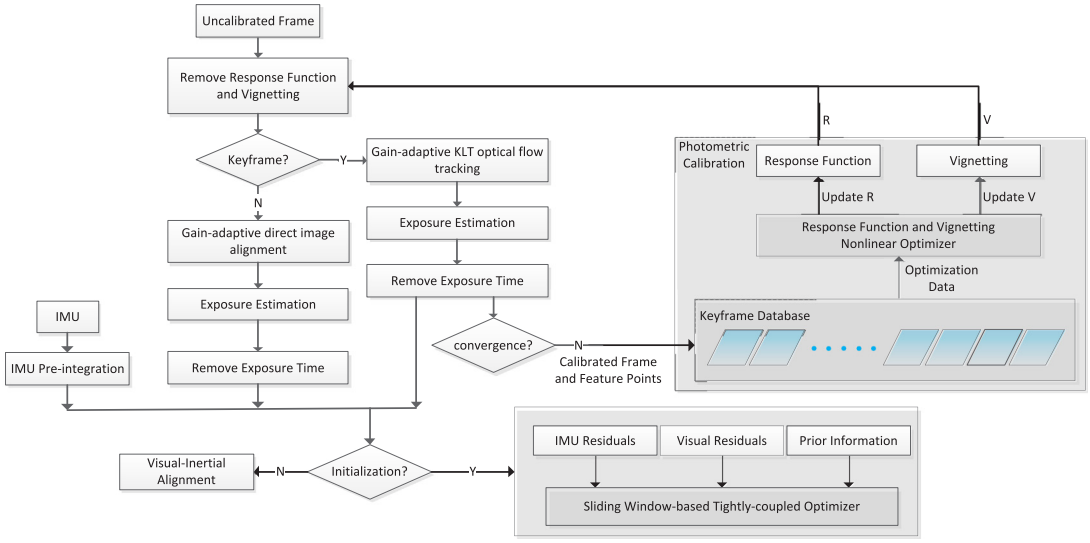


Fig. 1. Algorithmic overview of the system.

of the latest frame can be obtained by the exposure time ratio between adjacent frames, the estimation of exposure time is relatively rough because only two frames are considered. Therefore, after tracking the latest frame, we consider the keyframes in the keyframe database to estimate the exposure time of the latest frame and the radiance of the spatial feature points. Since the influence of the vignetting V and response function R has been eliminated, a series of feature points $p \in D$ on image $i \in F$ are obtained from the keyframe database, and we can obtain the following equation:

$$X = \arg\min_{(e_i, L^p)} \frac{1}{2} \sum_{i \in F} \sum_{p \in D} w_i^p \left\| \frac{R^{-1}\left(O_i^p\right)}{V\left(x_i^p\right)} - e_i L^p \right\|^2 \tag{4}$$

where O_i^p is the output intensity of p in image i, e_i is the exposure time of image i, L^p is the radiance of p and x_i^p is the projection position of the feature point p on the image i. w_i^p defines a weighting factor. $X = (e_i, L^p)$ is the optimization variable. By solving the above equation through the iterative Gauss–Newton method, the optimal solution of exposure time and feature point radiance can be obtained.

Different from [20], which considers all image frames, in order to be integrated into the SLAM algorithm, we only consider keyframes. If the latest frame is a keyframe, after eliminating the influence of the exposure time, the calibrated keyframe and feature points are stored in the keyframe database. The size of the keyframe database is 20. The photometric calibration module continuously obtains optimized data from the keyframe database and completes the continuous update of the response function R and vignetting V.

$$X = \arg\min_{(R,V)} \frac{1}{2} \sum_{i \in F} \sum_{p \in D} w_i^p \left\| O_i^p - R(V(x_i^p) L^p) \right\|^2 \tag{5}$$

where $X = (R, V)$ is the optimization variable. L^p is the radiance of p, and x_i^p is the projection position of the feature point p on the image i. w_i^p defines a weighting factor. O_i^p is the intensity of the feature point p on frame i. By solving the above equation through the iterative Gauss–Newton method, the optimal solution of the vignetting and response function can be obtained.

With the arrival of the latest frame, [20] continues to update the photometric parameters. However, when the feature points are unevenly distributed and the number of feature points of large radial motion is too small, the vignetting estimation value that has converged to the true value diverges. Unlike the work in [20], we propose a main criterion to determine whether the photometric calibration has converged and terminated. When the distribution frequency of feature points on the image exceeds 80% and the number of feature points with large radial motion in the keyframe database exceeds two-thirds of the total, we conclude that the estimation of the response function and vignetting has converged.

3.3. Tracking front-end based on the semi-direct method

Both photometric calibration and back-end optimization require sufficient feature points as inputs. Therefore, the main work of the front-end is to extract and match feature points and cooperate with the photometric calibration module to complete the calibration of the latest frame. We adopt a novel image tracking algorithm based on the semi-direct method. Non-keyframes are tracked through the direct method, and keyframes are tracked through the optical flow method, thus combining the accuracy of the feature-based method and the rapidity of the direct method.

Since we decouple the estimation of exposure time from other photometric parameters, the influence of the response function and vignetting is eliminated when the latest frame arrives. At this time, only the exposure time is not estimated. As shown in Fig. 2, considering that images taken by uncalibrated cameras may exhibit strong exposure changes, in these cases, the tracking effect of standard direct image alignment and KLT trackers is poor. Therefore, we propose gain-adaptive direct image alignment and gain-adaptive KLT optical flow tracking, which can jointly optimize the tracking of feature points and the exposure time ratio between adjacent frames [23]. Compared with the standard version, this can effectively deal with larger exposure time changes between images [24].

We adopt the gain-adaptive direct image alignment method to track non-keyframes. Since non-keyframes are only used for fast localization, feature point extraction is not needed. Keyframes are used for photometric calibration and back-end optimization, so feature point extraction is performed to ensure that we have a sufficient number of feature points, and the feature points are tracked through gain-adaptive KLT optical flow tracking technology. To ensure that the feature points can be evenly distributed on the image, we divide the image into several grid cells and extract a certain number of feature points in each sub-cell. We extract 5×5 image blocks around each new feature point so that

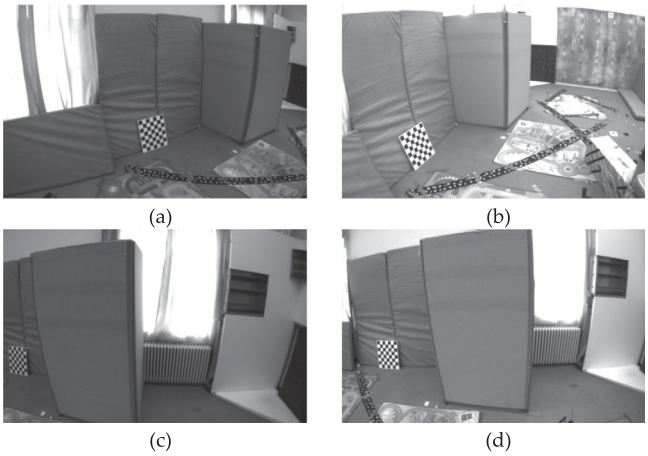


Fig. 2. Some sequences (a, b, c, d) in the EuRoC dataset have large exposure time changes.

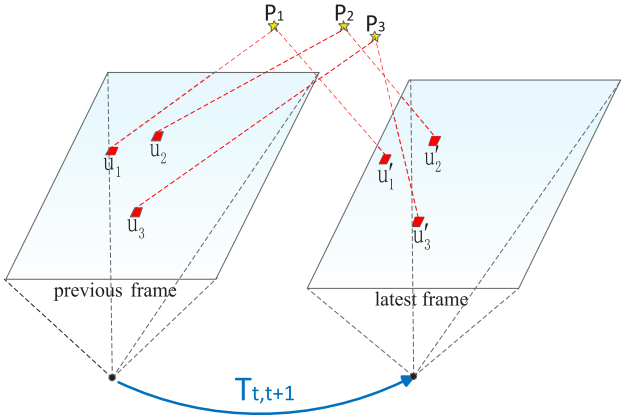


Fig. 3. Adjust the relative pose $T_{t,t+1}$ to minimize the photometric error of the image block on the latest frame and the previous frame.

more observations can be obtained. In addition, we also adopt the reverse optical flow tracking method to eliminate unstable points to ensure that there are more feature points with larger radial motion.

3.3.1. Gain-adaptive KLT optical flow tracking

Suppose that the same feature point can be observed by two adjacent frames I and J. Since the latest frame has eliminated the influence of the response function R and the vignetting V, the following equation can be obtained from the brightness constancy assumption.

$$\frac{R^{-1}(J(x))}{e_J V(x)} = \frac{R^{-1}(I(x+dx))}{e_I V(x+dx)} \tag{6}$$

Taking the logarithm of both sides of the equation, the following equation can be obtained.

$$\log\left(\frac{R^{-1}(J(x))}{V(x)}\right) - \log\left(\frac{R^{-1}(I(x+dx))}{V(x+dx)}\right) - \log\left(\frac{e_J}{e_I}\right) = 0 \tag{7}$$

Then, the following optimization equation can be obtained.

$$X = \arg\min_{(dx,K)} \frac{1}{2} \|f(J(x)) - f(I(x+dx)) - K\|^2 \tag{8}$$

where $X = (dx, K)$, $f(x) = \log(\frac{R^{-1}}{V(x)})$, and $K = \log\frac{e_J}{e_I}$. K is the logarithm of the exposure ratio between the two adjacent frames.J(x) and I(x) represent the intensity images in frames J and I, respectively. e_I and e_J are the exposure times of the adjacent frames I and J, respectively. $x = [x, y]^T$ is the feature point location, and $dx = [dx, dy]^T$ is the displacement vector.

Solving the optimization Equation (8), the displacement of $dx = [dx, dy]^T$ and the exposure difference K can be estimated.

3.3.2. Gain-adaptive direct image alignment

As shown in Fig. 3, to achieve adaptive direct image alignment, the relative pose $T_{t,t+1}$ between the latest frame and the previous frame should be roughly estimated [7]. The initial value of the relative pose can be obtained directly by IMU pre-integration. Since the influence of the response function and vignetting on the latest frame has been eliminated, only the exposure time affects the brightness constancy assumption at this time. We can jointly optimize the relative pose and the exposure time ratio between the two frames by minimizing the photometric error between the image blocks (5×5 pixels), thereby ensuring the validity of the brightness constancy assumption.

$$X = \arg\min_{(T_{t,t+1},K)} \frac{1}{2} \times \sum_{p\in P} \left\| g\left(I_{t+1}\left(\pi\left(T_{t,t+1}\cdot\pi^{-1}(u_p, d_p)\right)\right)\right) - g\left(I_t\left(u_p\right)\right) - K \right\|^2 \tag{9}$$

where $X = (T_{t,t+1}, K)$, $g(x) = \log(\frac{R^{-1}}{V(x)})$, and $K = \log\frac{e_{t+1}}{e_t}$. P represents all the feature points observed in the previous frame. e_t and e_{t+1} are the exposure times of the previous frame and the latest frame, respectively. d_p is the depth of the feature point in the previous frame. u_p are the projections of the feature points on the previous frame. I_k represents the intensity image in the *k*th frame. $\pi()$ is the projection equation. $\pi^{-1}()$ is the back projection equation. We adopt the iterative Gauss–Newton method to solve the above equation to obtain the relative pose $T_{t,t+1}$. As shown in Fig. 4, we define all 3D points observed in all frames in the sliding window as the local map [7]. By projecting the local map to the latest frame with a known relative pose, the projection point u'_p of the local map on the latest frame can be obtained. Since only two frames are considered, the estimation of the relative pose between the latest frame and the previous frame is relatively rough. This causes the projection point of the local map on the latest frame to be inaccurate, so the position of the projection point needs to be further optimized. The photometric error of the image block (5×5 pixels) is minimized in the latest frame and the reference frame to jointly optimize the position of the projection point and the exposure time ratio, thereby ensuring the validity of the brightness constancy assumption.

$$X = \arg\min_{(u'_p,K)} \frac{1}{2} \sum_{i\in S}\sum_{p\in M} \left\| h\left(I_{t+1}\left(u'_p\right)\right) - h\left(I_i\left(u_p\right)\right) - K \right\|^2 \tag{10}$$

where $X = (u'_p, K)$, $h(x) = \log(\frac{R^{-1}}{V(x)})$, and $K = \log\frac{e_{t+1}}{e_i}$. i represents the reference frame in sliding window S. e_i and e_{t+1} are the exposure times of the reference frame and the latest frame, respectively. p is a feature point in the local map M. u'_p and u_p are the projections of the feature points on the latest frame and the reference frame, respectively. I_k represents the intensity image in the *k*th frame. Solving the optimization Equation (10) by the iterative Gauss–Newton method, we can obtain a more accurate projection block position.

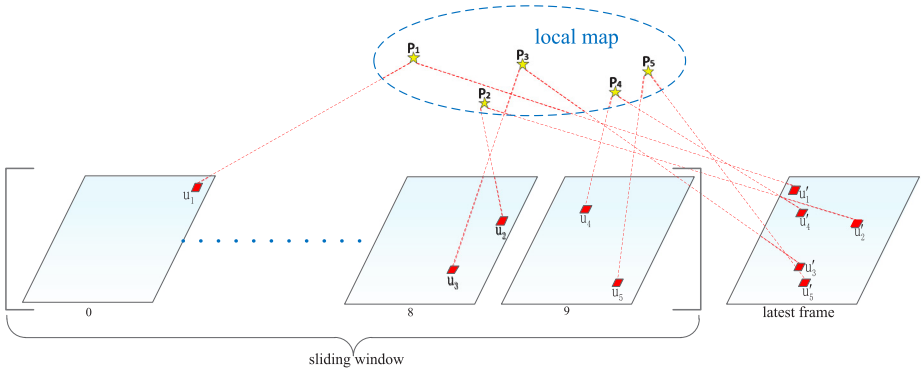


Fig. 4. Adjust the position of the projection block u'_p on the current frame to minimize the photometric error of the projection block in the current frame and the reference frame in the sliding window.

3.4. Sliding window-based monocular visual–inertial optimization framework

When performing image tracking and photometric calibration, the algorithm pre-integrates the IMU data at the same time [25]. The IMU pre-integration is aligned with the visual image during initialization, and the metric scale, initial velocity, gravity vector, and gyroscope bias are calibrated to complete the initialization of the system [5]. In the back-end optimization, the IMU pre-integration also participates in the state optimization as a constraint of the IMU between the two images.

After tracking the latest frame, it is optimized together with other frames in the sliding window. To ensure the real-time performance of the algorithm, a fixed number of keyframes are optimized in the sliding window, and the keyframes deleted from the sliding window are processed by marginalization. Our keyframe selection strategy determines whether the latest frame is a keyframe through the number of feature points, IMU pre-integration, and the average parallax of the feature points. If the latest frame is not a keyframe, it is deleted directly after the back-end optimization. If the latest frame is determined to be a keyframe, the frame remains in the sliding window after back-end optimization, and the oldest frame is marginalized. To ensure that the constraint information is not lost, the oldest frame is marginalized from the sliding window through the Schur complement. The new prior information constructed by the Schur complement is added to the existing prior information. The prior information generated by the marginalization is combined with the IMU residual error and the monocular reprojection error to jointly optimize the camera pose and feature point position to obtain high-precision localization and mapping [26].

The state variables to be estimated are defined as:

$$\begin{aligned} \mathrm{X} &= \begin{bmatrix} \mathrm{X}_0 & \mathrm{X}_1 & \dots & \mathrm{X}_n & \rho_0 & \rho_1 & \rho_2 \dots & \rho_m \end{bmatrix} \\ \mathrm{X}_k &= \begin{bmatrix} \mathrm{P}^{\mathrm{W}}_{\mathrm{B}_k} & \mathrm{V}^{\mathrm{W}}_{\mathrm{B}_k} & \mathrm{q}^{\mathrm{W}}_{\mathrm{B}_k} & \mathrm{b}^{\mathrm{B}}_{\alpha} & \mathrm{b}^{\mathrm{B}}_{\omega} \end{bmatrix} \quad \mathrm{k} \in [0, \mathrm{n}] \end{aligned} \tag{11}$$

where the state variables X are composed of the optimization variables X_n related to the IMU and feature point depths ρ_m, n is the size of the sliding window, and m is the number of feature points observed in the keyframe in the sliding window. B and W are the IMU body coordinate system and the world coordinate system, respectively. X_k includes the translation $\mathrm{P}^{\mathrm{W}}_{\mathrm{B}_k}$, velocity $\mathrm{V}^{\mathrm{W}}_{\mathrm{B}_k}$, and rotation quaternions $\mathrm{q}^{\mathrm{W}}_{\mathrm{B}_k}$, as well as the bias of gyroscope $\mathrm{b}^{\mathrm{B}}_{\alpha}$ and accelerometer $\mathrm{b}^{\mathrm{B}}_{\omega}$.

We optimize all state variables in the sliding window by minimizing the sum of all measured residuals:

$$\min_{\mathrm{X}} \left\{ \sum \|\mathrm{r}_{\mathrm{B}}\|^2 + \sum \|\mathrm{r}_{\mathrm{C}}\|^2 + \left\|\mathrm{r}_{\mathrm{p}}\right\|^2 \right\} \tag{12}$$

where r_{B}, r_{C} and r_{p} are IMU residuals, visual reprojection errors and prior information, respectively.

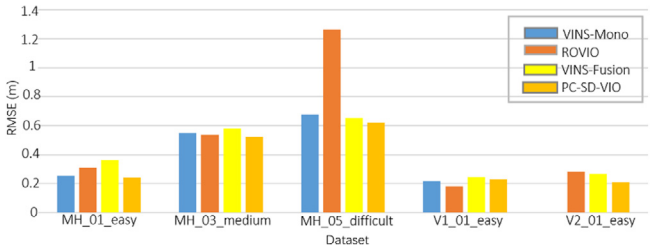


Fig. 5. RMSE for VINS-Mono, ROVIO, VINS-Fusion, and PC-SD-VIO in the EuRoC dataset.

4. Experiments

We evaluate the performance of PC-SD-VIO on the EuRoC dataset and the TUM Mono dataset [27]. Since the EuRoC dataset can only provide the ground truth trajectory, but not the real photometric parameter values, we mainly evaluate the accuracy and real-time performance of the algorithm on the EuRoC dataset. The TUM Mono VO dataset can obtain real photometric parameter values, so we adopt the TUM Mono VO dataset to evaluate the effectiveness and accuracy of the online photometric calibration module. Finally, we verified the robustness and feasibility of the algorithm and the photometric calibration module through physical experiments in real environments. The above experiments were performed on the laptop configured with 2.80 GHz, Intel Core i7-7700HQ CPU, 16 GB RAM, and ROS Kinetic.

In the EuRoC dataset, we compare PC-SD-VIO with the state-of-the-art monocular SLAM algorithms VINS-Mono, ROVIO, and VINS-Fusion to validate the advantages and disadvantages of the algorithm [28]. As shown in Table 1, we use the root mean square error (RMSE) of the trajectory error as the index to evaluate the accuracy of the algorithm and the average time required to track each frame as the index to evaluate the real-time performance of the algorithm.

As seen in Table 1 and Fig. 5, PC-SD-VIO is superior to other algorithms in terms of accuracy and robustness, which benefits from the addition of an online photometric calibration module. Although the direct method and optical flow method are susceptible to brightness instability when tracking image frames, our front-end can coordinate with the photometric calibration module to calibrate the latest frame in time, thus ensuring the validity of the brightness constancy assumption. As shown in Fig. 6, we observed that in some datasets, there were strong exposure time changes between images. In these cases, the tracking accuracy of the standard optical flow method and the standard direct method is reduced or even fails. This caused VINS-Mono

Table 1
The root mean square error (RMSE) of trajectory error (m) and the average time (ms) required to track the image.

Dataset	VINS-Mono		ROVIO		VINS-Fusion		PC-SD-VIO	
	Time(ms)	RMSE(m)	Time(ms)	RMSE(m)	Time(ms)	RMSE(m)	Time(ms)	RMSE(m)
MH_01_easy	17.7	0.255	12.9	0.309	38.9	0.364	**6.71**	**0.244**
MH_03_medium	12.4	0.548	10.1	0.537	44.2	0.582	**7.85**	**0.522**
MH_05_difficult	12.8	0.678	9.81	1.27	42.7	0.654	**6.93**	**0.621**
V1_01_easy	13.4	0.217	10.5	**0.182**	45.9	0.247	**7.24**	0.232
V2_01_easy	×	×	11.9	0.283	49.1	0.267	**6.38**	**0.212**

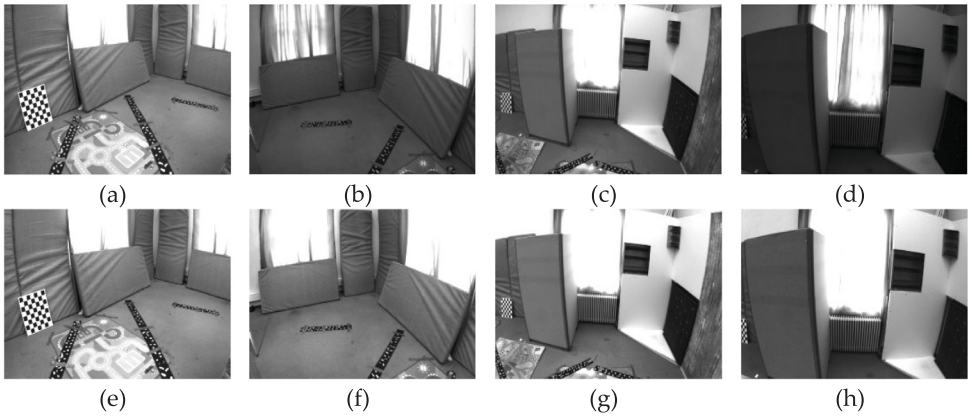


Fig. 6. The top (a, b, c, d) shows the image without photometric calibration in the EuRoC dataset. The bottom (e, f, g, h) shows the calibrated image.

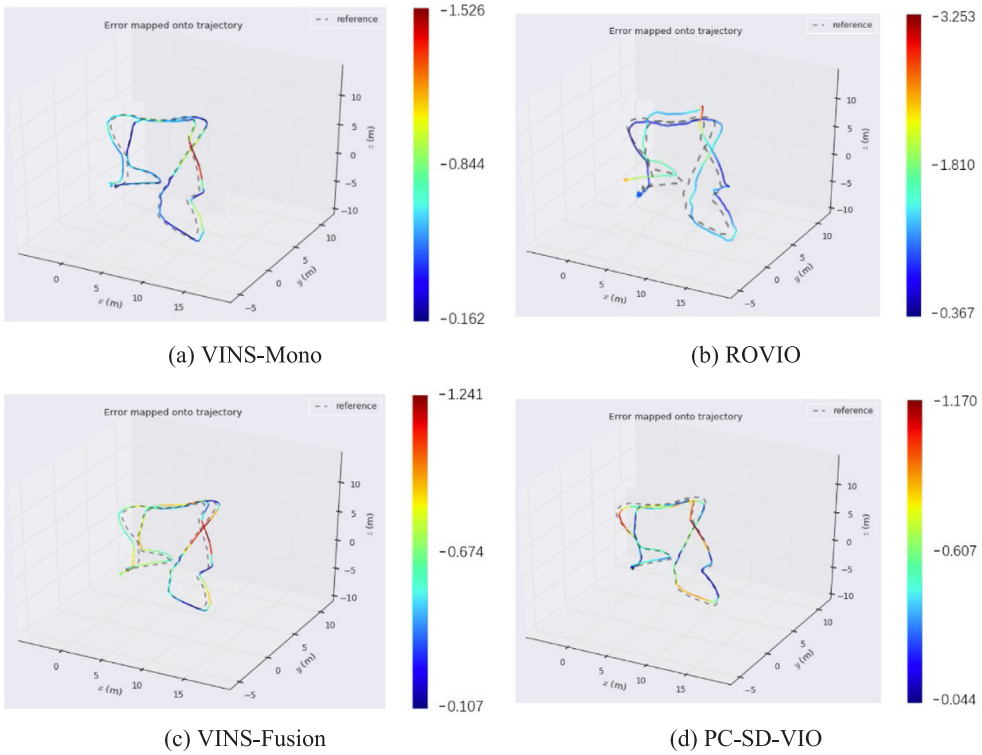
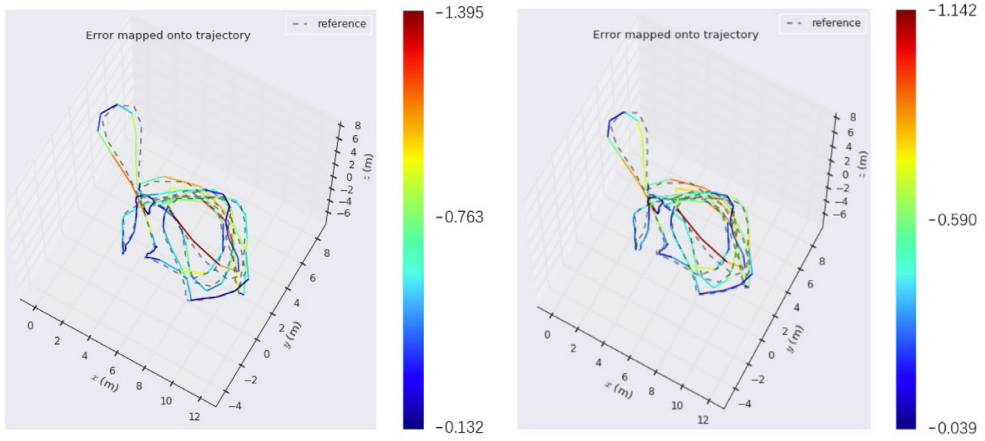


Fig. 7. The trajectory heat map estimated by VINS-Mono, ROVIO, VINS-Fusion, and PC-SD-VIO in MH_05_difficult.

to fail to run on the V2_01_easy dataset. However, the front-end of PC-SD-VIO combines gain-adaptive direct image alignment and gain-adaptive KLT optical flow tracking, which can effectively deal with large exposure time changes between adjacent images, so it can run successfully on the V2_01_easy dataset. In addition, due to the combination of optical flow and the direct method used in the front-end to track and process images, the algorithm has greater advantages in low-texture environments such as V2_01_easy. Fig. 7 more intuitively shows the trajectory heat map estimated by VINS-Mono, ROVIO, VINS-Fusion, and PC-SD-VIO on MH_05_difficult.

To further verify the influence of photometric calibration on the SLAM algorithm, we compared the proposed algorithm with and without photometric calibration on the MH_03_medium dataset. As shown in Fig. 8, the PC-SD-VIO algorithm with photometric calibration achieved results with better accuracy. This



(a) Without photometric calibration (b) With photometric calibration

Fig. 8. The trajectory heat map estimated on MH_03_medium by PC-SD-VIO with and without photometric calibration.

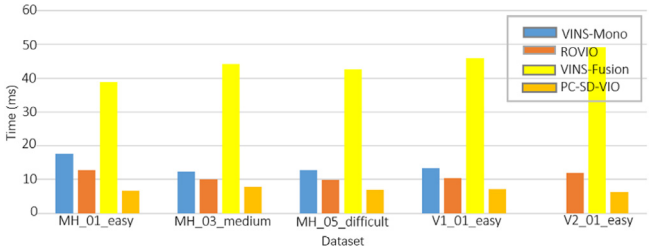


Fig. 9. The average time required to track each frame in VINS-Mono, ROVIO, VINS-Fusion, and PC-SD-VIO.

Table 2
The average time consumed by PC-SD-VIO to execute each module.

Module	Time consumption
Front-end tracking	7.85 ms
Exposure time	1.03 ms
Vignetting and response function	11526.48 ms
Back-end optimization	22.97 ms

further proves that with the integration of the photometric calibration module, the accuracy of the SLAM algorithm can be significantly improved.

It is worth noting that in the V1_01_easy dataset, PC-SD-VIO performs worse than VINS-Mono and ROVIO. We found that after completing the photometric calibration of the image, the dark areas become darker, and the bright areas become brighter. In other words, it changes the image intensity histogram distribution and reduces the image contrast. Since the extraction and tracking of feature points are usually better on images with higher contrast, the photometric calibration may have a negative impact on the SLAM algorithm. The front-end of the four algorithms participating in the experiment are all set to extract at most 200 feature points on each frame of the image. The real-time performance of these algorithms is evaluated by the average time required to track a frame of an image. As seen in Table 1 and Fig. 9, in terms of real-time performance, PC-SD-VIO is superior to the other algorithms. The reason is that the optical flow algorithm is used to track the keyframes, thereby eliminating the calculation process of the feature point descriptor. For a large number of non-keyframes, only the direct method is used to track existing feature points, and new feature points are not extracted. For the added online photometric calibration module, since the response function and vignetting are optimized in the background, the photometric calibration does not affect the real-time performance of the algorithm. Fig. 10 shows the photometric parameters estimated by our algorithm on the MH_03_medium dataset. Since the EuRoC dataset does not provide real photometric parameter values, the accuracy of photometric estimation has not been evaluated.

Table 2 shows the average time consumed by each module when our algorithm runs on the MH_03_medium dataset. It takes 31.85 ms to complete the front-end tracking, exposure time estimation, and back-end optimization, which shows that our algorithm can process images in real-time at a frequency of approximately 31HZ. The keyframe database in the photometric calibration module stores 20 keyframes, and 200 feature points are extracted from each frame, which results in a slow estimation of the vignetting and response function, with an average time of 11.5 s. However, since the photometric calibration module can be executed in the background, it does not negatively affect the real-time performance of the entire algorithm.

We also compared our algorithm with that of [20] on the TUM Mono VO dataset, which can provide real photometric parameters, so the feasibility and accuracy of the photometric calibration module can be evaluated. Fig. 11 shows the estimated photometric parameters when running on sequence 50. Our online photometric calibration module and that of [20] are basically consistent with the true value in the estimation of the response function and exposure time.

However, the estimation effect of the work in [20] on vignetting is not ideal. In our experiments, we found that the converged vignetting parameters (the green curve in Fig. 11b) are extremely susceptible to the influence of feature points with uneven distribution and small radial motion, which causes the converged vignetting parameters to diverge again (the orange curve in Fig. 11b). The fundamental reason is that the response function and vignetting parameters are constantly updated in [20]. To solve this problem, we propose a main criterion to determine whether the photometric calibration has converged and terminated. When the distribution frequency of feature points on the image exceeds 80% and the number of feature points with large radial motion in the keyframe database exceeds two-thirds of the total, we stop the estimation of the response function and vignetting. Therefore, our algorithm achieved better results in the estimation of vignetting parameters.

To verify the robustness and feasibility of the algorithm and the photometric calibration module in real experiments, we conducted physical experiments in the laboratory with an Intel Realsense D435i camera. The output trajectory is shown in Fig. 12.

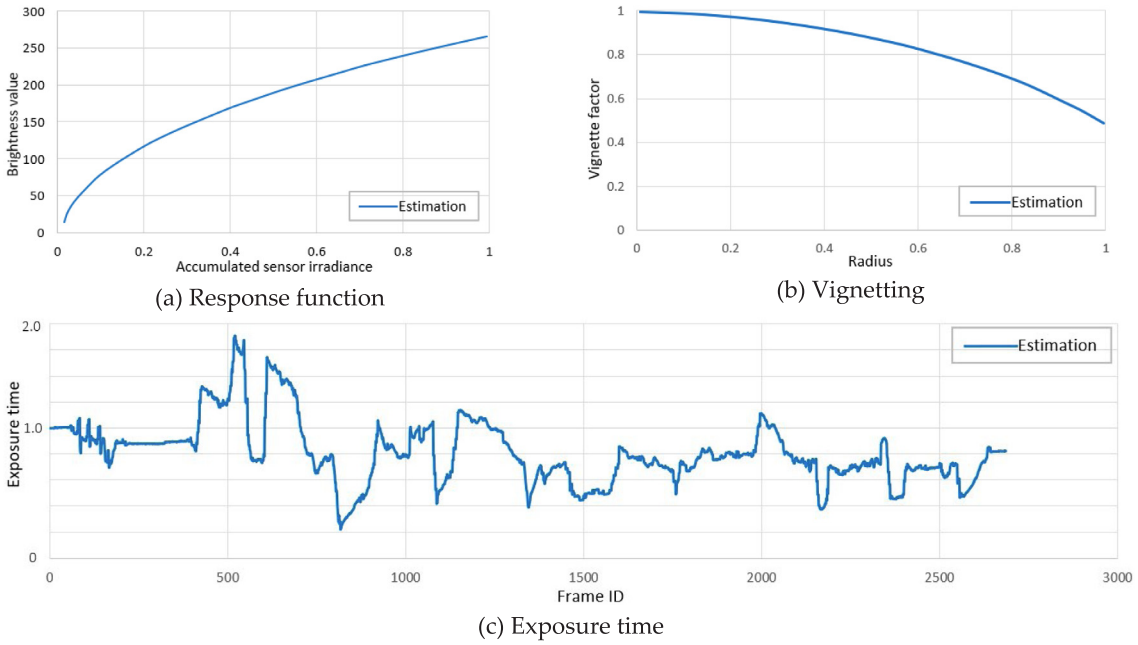


Fig. 10. The photometric parameters estimated by PC-SD-VIO in MH_03_medium.

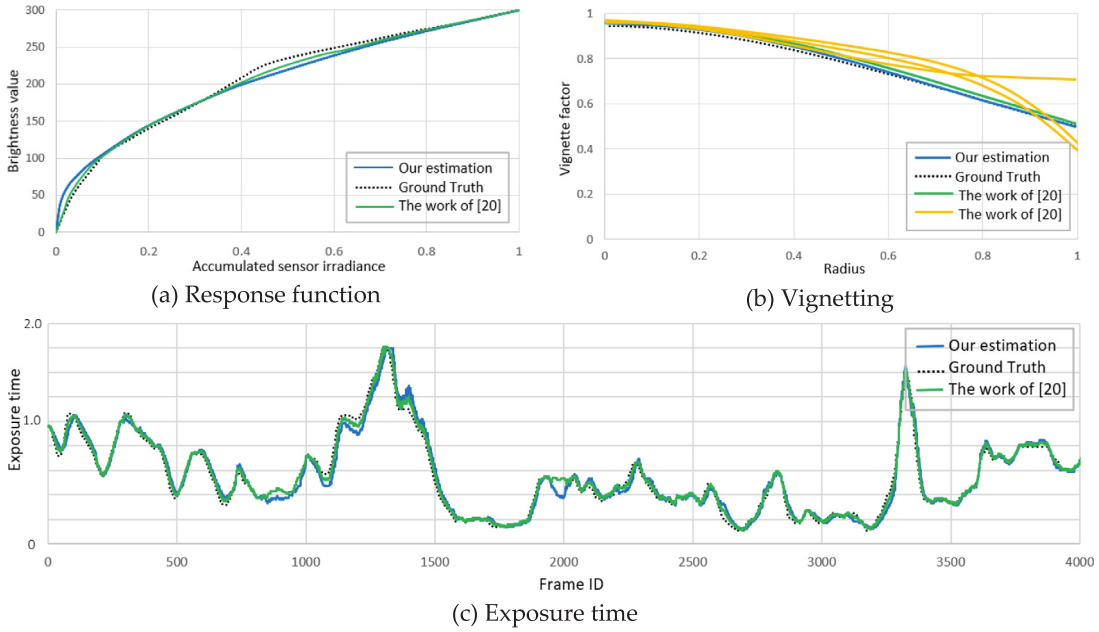


Fig. 11. The photometric parameters estimated by PC-SD-VIO and [20] in sequence 50 of the TUM Mono VO dataset.

Since the ground truth trajectory cannot be obtained, only the robustness and feasibility of the algorithm are verified here. Fig. 13 shows the photometric parameters estimated by the photometric calibration module in real experiments. Fig. 14 shows the comparison between the original image and the calibrated image. It can be seen that the camera automatically adjusts the exposure time in a laboratory environment with uneven lighting, thus breaking the assumption of constant brightness. In this case, without the support of the photometric calibration module, it is difficult for the SLAM algorithm to achieve high-precision performance, and it might even fail. However, after the video sequence with large changes in exposure time is processed by the online photometric correction module, the negative effects of exposure time, vignetting, and response function can be effectively eliminated. Therefore, our algorithm can achieve good performance in complex lighting environments. In Fig. 14, we also note that after completing the photometric calibration of the image, the bright areas become brighter, thereby reducing the image contrast. Therefore, the photometric calibration may have some negative effects on the SLAM algorithm.

5. Conclusion

We propose a novel semi-direct VIO system with an online photometric calibration module that can estimate the complete photometric parameters of arbitrary video sequences and uncalibrated cameras. To integrate with the photometric calibration

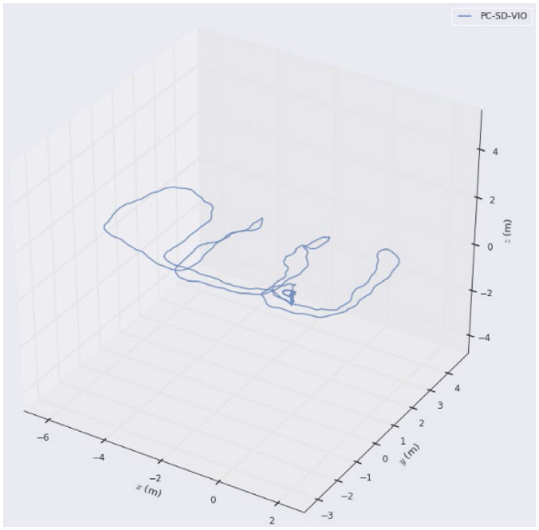


Fig. 12. The trajectory of physics experiment.

module, we have improved the standard version of direct image alignment and KLT optical flow tracking so that it can simultaneously complete the tracking of feature points and the estimation of the exposure time ratio between adjacent frames. We have achieved a better balance between accuracy, speed and robustness, so this algorithm is more suitable for platforms with limited computing resources in real environments. In the future, we will consider integrating the online photometric calibration module into the existing direct method SLAM to improve system robustness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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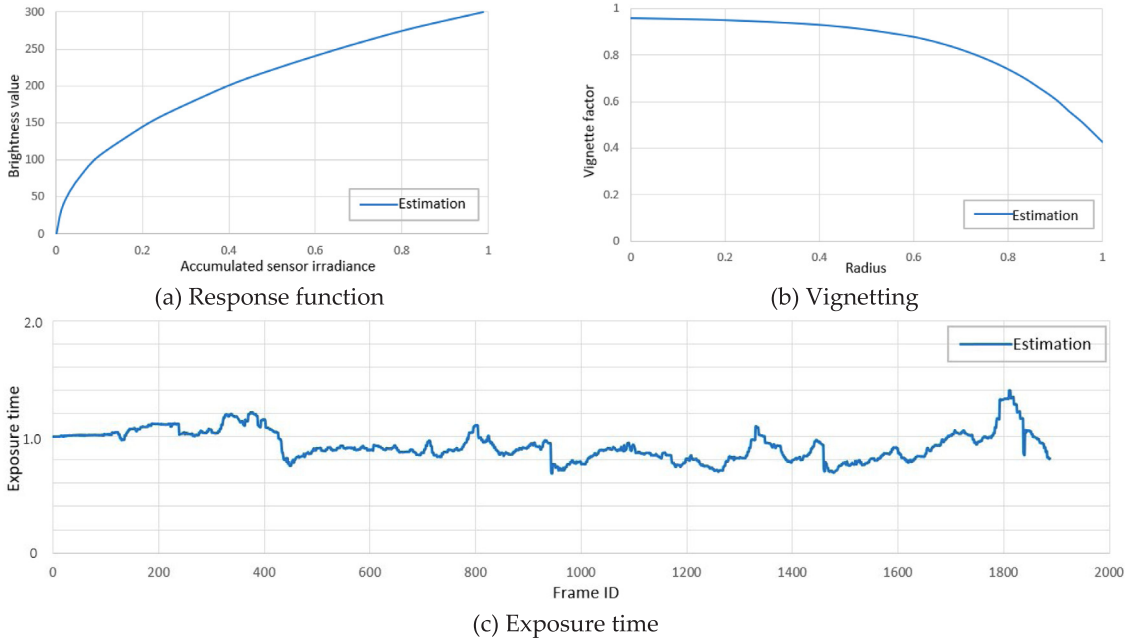


Fig. 13. The photometric parameters estimated by PC-SD-VIO in real experiments.

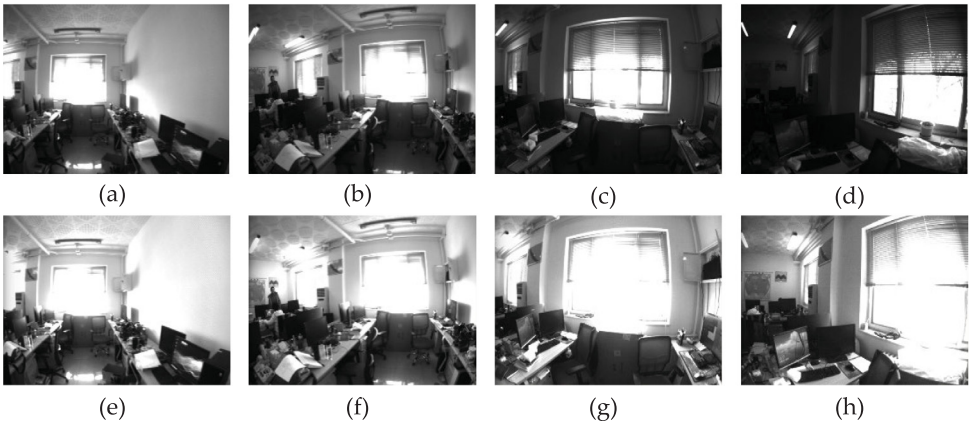


Fig. 14. The top (a, b, c, d) shows the image without photometric calibration in laboratory environments. The bottom (e, f, g, h) shows the calibrated image.

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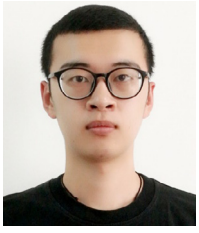
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